

3.4 SLOPE CONTROLS OPTIMIZATION ACCESS TO HIGH PRECISION

Section 3.3 identifies a useful relative bandwidth $\lambda = \gamma h$ for the construction. Holding it fixed as width increases requires $\gamma = \Theta(W)$. Training must acquire a useful feature geometry as well as fit its readout; the latter can remain slow even when the features already represent the target accurately. We derive a slope-dependent lower bound on output error for the finite tanh construction. The underlying scale-dependent spectral bias is established prior work (Xu et al., 2020; Tancik et al., 2020).

Slope controls access to representable corrections. Freeze the uniform centers from Section 3.1 and a common slope γ . On observations z_a , let J_γ contain the tanh features and output bias, divided by $\sqrt{m}$, and let $K_\gamma = J_\gamma J_\gamma^T$. Write its descending eigenpairs as (μ_i, u_i) , its relative rates as $\rho_i = \mu_i/\mu_1$, and the target weights as $p_i = |u_i^T y|^2/\|y\|^2$, where $y_a = f(z_a)/\sqrt{m}$. The largest eigenvalue limits a stable shared GD step; the ratios ρ_i determine how quickly each target component can then decay.

The localized derivative in Section 3.1 also explains the optimization effect: a tanh is a step smoothed by $s_\gamma(x) = \frac{\gamma}{2} \text{sech}^2(\gamma x)$. This smoothing multiplies frequency ω by

$$M_\gamma(\omega) = \frac{\pi|\omega|/(2\gamma)}{\sinh(\pi|\omega|/(2\gamma))}, \quad M_\gamma(0) = 1. \quad (9)$$

Smaller slopes attenuate rapid variation more strongly. Integrating these features over centers at density $1/h$ gives a kernel whose directional response is weighted by M_γ^2 . For an orthonormal basis Q of mean-zero sample directions, this integral is explicitly

$$H_\gamma = -\frac{2}{hm} Q^T [d_{ab} \coth(\gamma d_{ab})]_{a,b} Q, \quad d_{ab} = z_a - z_b, \quad (10)$$

with diagonal value $1/\gamma$ inside brackets. The integral exposes how gamma filters each sample direction; finite-center corrections transfer this calculation to the dictionary actually trained. Specifically, subtract an explicit contribution T_γ from exterior lattice centers absent from that dictionary, giving $S_\gamma = H_\gamma - T_\gamma$. Poisson summation supplies a lattice-error allowance δ_γ such that $Q^T K_\gamma Q \preceq S_\gamma + \delta_\gamma I$. Both corrections are defined in Appendix A.

Theorem 3.2 (Slope-dependent output-error lower bound). *Let consecutive uniformly spaced centers cover the sample interval, with common slope $\gamma > 0$, and let $y \neq 0$. Let $\beta_i(\gamma)$ be the descending eigenvalues of S_γ above and $\ell_\gamma = v_0^T K_\gamma v_0$ for $v_0 = \mathbf{1}/\sqrt{m}$. Define*

$$\bar{\rho}_1 = 1, \quad \bar{\rho}_i(\gamma) = \min \left\{ 1, \frac{\beta_{i-1}(\gamma) + \delta_\gamma}{\ell_\gamma} \right\}, \quad 2 \leq i \leq m. \quad (11)$$

Gradient descent on $\frac{1}{2}\|J_\gamma w - y\|^2$ from $w_0 = 0$, with $\eta = 1/(2\mu_1)$, satisfies, for every integer $n \geq 0$,

$$\boxed{\frac{\|J_\gamma w_n - y\|^2}{\|y\|^2} \geq \sum_i p_i(\gamma) \left(1 - \frac{\bar{\rho}_i(\gamma)}{2} \right)^{2n}}. \quad (12)$$

If directions carrying target energy P have rate upper bounds at most r , they force relative output error of at least $\sqrt{P}(1 - r/2)^n$. This includes positive-eigenvalue components that are representable but slow to acquire.

Interlacing gives $\rho_i \leq \bar{\rho}_i$; substituting into GD's exact squared relative error $\sum_i p_i(1 - \rho_i/2)^{2n}$ proves the bound (Appendix A). Target projections use the actual dictionary; the rate bounds come from the slope-dependent integral and finite-center corrections before training.

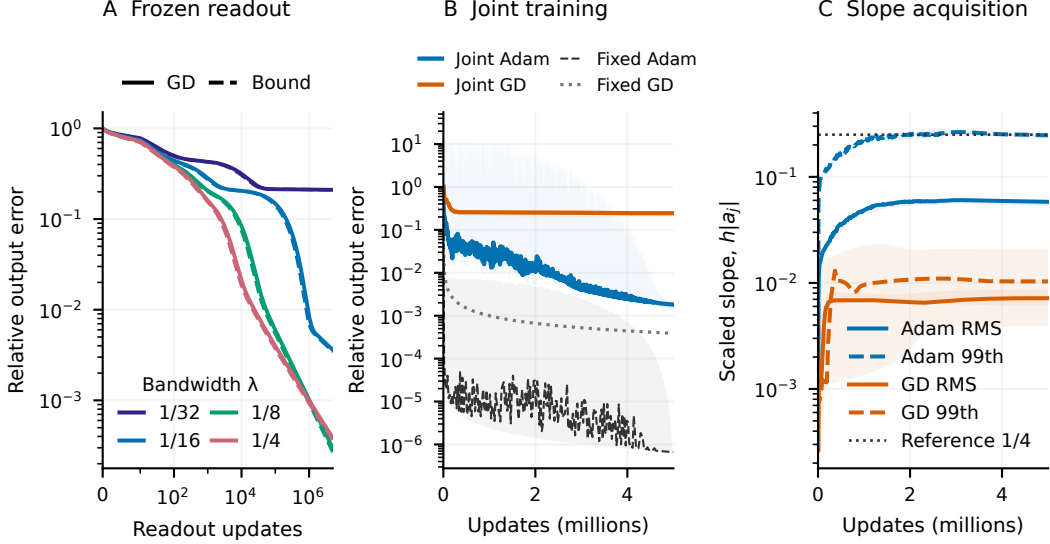


Figure 3: **Slope affects access to representable precision.** All panels use $[\sin(2\pi x) + \frac{1}{2}\sin(6\pi x) + \frac{1}{4}\sin(14\pi x)]/\sqrt{21/32}$, hidden width 512 including halos, 2,048 training midpoints, and five million updates. **A:** actual frozen-readout GD errors (solid) and square roots of the theorem’s error lower bounds (dashed), at $\eta = 1/(2\mu_1)$. **B:** joint Adam/GD errors, with uniform $\lambda = 1/4$ frozen-feature references. Each joint optimizer uses one recipe selected by median final validation error across five paired seeds. **C:** RMS and 99th-percentile $h|a_j|$ from exactly B’s joint runs. The uniform reference is a comparison, not a necessary threshold for heterogeneous features. Lines summarize raw errors within display bins; shading retains seed variation and within-bin extrema, including spikes. Cosine decay spans the full run. The appendix gives recipes, validation, and numerical checks.

Accuracy can be representable but slow to acquire. Figure 3A isolates the readout effect on a fixed width-512 geometry. At $\lambda = 1/32$, a direct readout fit attains 6.2×10^{-10} relative error on the dense evaluation grid, while GD retains training-sample error 0.210 after five million updates; the theorem predicts at least 0.2098. At $\lambda = 1/4$, the same GD protocol reaches 3.88×10^{-4} . Across all four bandwidths and every executed update, the theorem lower bound is at least 75% of the actual error, and roughly 91–100% at the endpoint. The smoothing mechanism identifies the difficult correction: increasing λ from 1/32 to 1/4 raises the feature multiplier at the target’s highest frequency from 0.00146 to 0.800, and that harmonic accounts for 95.4% of the small-slope GD residual energy at five million updates. Thus the bound quantifies delayed access to accuracy already available in the dictionary.

Joint training acquires useful features, but precision remains costly. Joint Adam reduces median relative error to 1.81×10^{-3} , while Adam applied only to the readout of the supplied $\lambda = 1/4$ dictionary reaches 6.62×10^{-7} over the same horizon. Joint GD remains at 0.244 (Figure 3B). For jointly trained features $\tanh(a_j x + b_j)$, Figure 3C tracks scaled slopes $h|a_j|$. Adam’s endpoint median RMS is 0.0582, and its 99th percentile approaches 1/4, but only 4–7 of the 512 features reach that scale. A large upper percentile therefore does not mean the uniform reference geometry has been acquired.

Longer training helps: Adam’s median endpoint error falls 28% between the two- and five-million-update experiments and continues improving late in the latter. Yet checked FP64 direct fits to the learned five-million-update features barely improve their median error. Separate interventions at two million updates show that steepening those features or replacing their centers alone can worsen accuracy (Appendix B). The joint-training gap therefore concerns acquisition of a useful combination

of slopes and centers; the frozen theorem isolates how slope controls optimization once centers are fixed.

The construction supplies accurate feature geometry and readout weights directly. It therefore avoids both the search for that geometry and slow iterative access to its fine corrections. This motivates composing the constructed primitives into arithmetic circuits and placing those computations inside larger networks.

A Proof of the slope-dependent output-error bound

The continuous-center integral exposes how the common tanh slope changes the readout kernel. An explicit lattice allowance transfers this calculation to finitely many equally spaced centers. Interlacing supplies upper bounds on the relative learning rates; substituting these bounds into the exact GD residual formula gives the output-error lower bound. We separate this analytic argument from its numerical evaluation below.

Setup and statement. Fix distinct samples $z_1, \dots, z_m$ and W consecutive centers x_j on $c_0 + h\mathbb{Z}$, with $h > 0$, covering the sample interval. Fix $\gamma > 0$ and set

$$J_\gamma = \frac{1}{\sqrt{m}} [\mathbf{1}, (\tanh(\gamma(z_a - x_j)))_{a,j}], \quad K_\gamma = J_\gamma J_\gamma^T, \quad y_a = \frac{f(z_a)}{\sqrt{m}}. \quad (\text{A.1})$$

Thus $\|J_\gamma w - y\|^2$ is the mean squared sampled output error. Assume $y \neq 0$, write the descending eigenpairs of K_γ as $(\mu_i, u_i)_{i=1}^m$, and define

$$\rho_i = \mu_i / \mu_1, \quad p_i = |u_i^T y|^2 / \|y\|^2, \quad v_0 = \mathbf{1} / \sqrt{m}. \quad (\text{A.2})$$

The weights sum to one, including energy in the exact nullspace. The symbol $\lambda = \gamma h$ continues to denote relative bandwidth, not an eigenvalue.

Let Q have orthonormal columns spanning $v_0^\perp$, and let $t_\gamma(c) = (\tanh(\gamma(z_a - c)))_a$. Define

$$H_\gamma = \frac{1}{hm} \int_{\mathbb{R}} Q^T t_\gamma(c) t_\gamma(c)^T Q dc = -\frac{2}{hm} Q^T [d_{ab} \coth(\gamma d_{ab})]_{a,b} Q, \quad d_{ab} = z_a - z_b, \quad (\text{A.3})$$

where the diagonal value inside brackets is $1/\gamma$. Retain a finite set $\mathcal{F}$ of absent exterior lattice centers such that the remaining absent centers form two tails, starting at $c_L < z_{\min}$ and $c_R > z_{\max}$. Put

$$T_\gamma = \frac{1}{m} \sum_{c \in \mathcal{F}} Q^T t_\gamma(c) t_\gamma(c)^T Q, \quad S_\gamma = H_\gamma - T_\gamma, \quad (\text{A.4})$$

and let $\beta_1 \geq \dots \geq \beta_{m-1}$ be the eigenvalues of S_γ . For any $0 < \vartheta < \pi/2$, define

$$\delta_\gamma = \frac{8\gamma \|z - \bar{z}\mathbf{1}\|^2 \sec^4 \vartheta}{3hm [\exp(2\pi\vartheta/(\gamma h)) - 1]}, \quad \bar{z} = m^{-1} \sum_a z_a, \quad \ell_\gamma = v_0^T K_\gamma v_0. \quad (\text{A.5})$$

Output-error lower bound. Set $\bar{\rho}_1 = 1$ and

$$\bar{\rho}_i(\gamma) = \min \left\{ 1, \frac{\beta_{i-1}(\gamma) + \delta_\gamma}{\ell_\gamma} \right\}, \quad 2 \leq i \leq m. \quad (\text{A.6})$$

Then $\rho_i \leq \bar{\rho}_i \leq 1$. Gradient descent from $w_0 = 0$ on $\frac{1}{2} \|J_\gamma w - y\|^2$, with $\eta = 1/(2\mu_1)$, satisfies

$$\frac{\|J_\gamma w_n - y\|^2}{\|y\|^2} \geq \sum_{i=1}^m p_i(\gamma) \left(1 - \frac{\bar{\rho}_i(\gamma)}{2} \right)^{2n} \quad (n \geq 0). \quad (\text{A.7})$$

Known exact zero eigenvalues may have their endpoints set to zero; in particular, $\text{rank } K_\gamma \leq W + 1$. The conclusion depends on target energy in the bounded slow directions; it does not assert that every target is slow at small slope.

A.1 THE CONTINUOUS-CENTER KERNEL EXPOSES GAMMA

Removing the constant sample pattern makes the integral finite because $Q^T t_\gamma(c)$ decays exponentially at both ends. The identity $1 - \tanh u \tanh v = \coth(u - v)(\tanh u - \tanh v)$ gives

$$\int_{\mathbb{R}} [1 - \tanh(\gamma(z_a - c)) \tanh(\gamma(z_b - c))] dc = 2d_{ab} \coth(\gamma d_{ab}), \quad (\text{A.8})$$

with diagonal limit $2/\gamma$. Multiplication by Q^T and Q cancels the constant term and proves Eq. (A.3).

Write $s_\gamma(t) = \frac{\gamma}{2} \text{sech}^2(\gamma t)$. The identity $\tanh(\gamma \cdot) = s_\gamma * \text{sign}$ follows because both sides vanish at zero and have derivative $2s_\gamma$. For the convention $\hat{g}(\omega) = \int g(t) e^{-i\omega t} dt$, substitution $r = e^{2\gamma t}$, the beta integral, and the gamma-function reflection formula yield

$$\begin{aligned} \hat{s}_\gamma(\omega) &= \int_0^\infty \frac{r^{-i\omega/(2\gamma)}}{(1+r)^2} dr = \Gamma\left(1 - \frac{i\omega}{2\gamma}\right) \Gamma\left(1 + \frac{i\omega}{2\gamma}\right) \\ &= \frac{\pi\omega/(2\gamma)}{\sinh(\pi\omega/(2\gamma))} =: M_\gamma(\omega), \quad M_\gamma(0) = 1. \end{aligned} \quad (\text{A.9})$$

For $u = Qv$, the step combination $g_0(c) = \sum_a u_a \text{sign}(c - z_a)$ has compact support since $\sum_a u_a = 0$. Its distributional derivative is $2 \sum_a u_a \delta_{z_a}$, so $\hat{g}_0(\omega) = 2 \sum_a u_a e^{-i\omega z_a} / (i\omega)$. Parseval therefore gives

$$v^T H_\gamma v = \frac{2}{\pi h m} \int_{\mathbb{R}} \frac{M_\gamma(\omega)^2}{\omega^2} \left| \sum_a (Qv)_a e^{-i\omega z_a} \right|^2 d\omega. \quad (\text{A.10})$$

The zero-frequency singularity is removable because the numerator vanishes there. The direction's coefficients $(Qv)_a$ determine its frequency content; gamma changes its weight through M_γ^2 . Since $d \log(z / \sinh z) / dz = 1/z - \coth z < 0$ for $z > 0$, increasing gamma increases M_γ at every fixed frequency. Thus $H_\Gamma \succeq H_\gamma$ for $\Gamma \geq \gamma$, and min-max gives ordered growth of the integral eigenvalues. The eigenvectors may change. Exterior-center subtraction and normalization by the largest finite eigenvalue need not preserve this order.

A.2 TRANSFER TO A FINITE CENTER LATTICE

For $u \perp \mathbf{1}$, let $g(\zeta) = \sum_a u_a \tanh(\gamma(\zeta - z_a))$, analytic in $|\text{Im } \zeta| < \pi/(2\gamma)$. Subtract the common translate at $\bar{z}$, write each difference as an integral of its derivative, and apply Minkowski and Cauchy-Schwarz. On the strip boundaries this gives

$$\int_{\mathbb{R}} |g(t \pm i\vartheta/\gamma)|^2 dt \leq \frac{4\gamma}{3} \sec^4 \vartheta \|z - \bar{z}\|^2 \|u\|^2. \quad (\text{A.11})$$

Indeed, $|\text{sech}^2(t + i\vartheta)| \leq \sec^2 \vartheta \text{sech}^2 t$ and $\int \text{sech}^4 t dt = 4/3$, so the L^2 norm of a translate difference is at most $|z_a - \bar{z}| \sqrt{4\gamma/3} \sec^2 \vartheta$.

Shift the contour for the analytic function $g(\zeta)^2$, not $|g(\zeta)|^2$. Its transform is bounded by Eq. (A.11) times $e^{-\vartheta|\omega|/\gamma}$. Poisson summation on $c_0 + h\mathbb{Z}$ and the geometric sum over frequencies $2\pi k/h$, $k \neq 0$, imply

$$\left\| \frac{1}{m} \sum_{c \in c_0 + h\mathbb{Z}} Q^T t_\gamma(c) t_\gamma(c)^T Q - H_\gamma \right\| \leq \delta_\gamma. \quad (\text{A.12})$$

The lattice origin enters only through phases of modulus one.

The finite kernel removes all absent centers. Let T_{tail} contain the absent centers not retained in $\mathcal{F}$. Since $Q^T \mathbf{1} = 0$ and $1 - \tanh t \leq 2e^{-2t}$ for $t \geq 0$, a right-tail projected outer product divided by m has norm at most $4e^{-4\gamma(c_R - z_{\max})}$; the left tail is analogous. Hence

$$0 \preceq T_{\text{tail}} \preceq \tau_\gamma I, \quad \tau_\gamma = \frac{4[e^{-4\gamma(c_R - z_{\max})} + e^{-4\gamma(z_{\min} - c_L)}]}{1 - e^{-4\gamma h}}. \quad (\text{A.13})$$

Writing the lattice discrepancy as R_{lattice} yields

$$Q^T K_\gamma Q = S_\gamma + R_{\text{lattice}} - T_{\text{tail}}, \quad \|R_{\text{lattice}}\| \leq \delta_\gamma, \quad (\text{A.14})$$

and consequently

$$S_\gamma - (\delta_\gamma + \tau_\gamma)I \preceq Q^T K_\gamma Q \preceq S_\gamma + \delta_\gamma I. \quad (\text{A.15})$$

The bias vanishes under Q . The unretained exterior tail can only lower the finite kernel; it is unnecessary for its upper eigenvalue bound.

A.3 RELATIVE RATES AND GD OUTPUT ERROR

If κ_i are the descending eigenvalues of $Q^T K_\gamma Q$, min-max and codimension-one interlacing give

$$\beta_i - \delta_\gamma - \tau_\gamma \leq \kappa_i \leq \beta_i + \delta_\gamma, \quad \mu_i \geq \kappa_i \geq \mu_{i+1}. \quad (\text{A.16})$$

Thus $\mu_i \leq \beta_{i-1} + \delta_\gamma$ for $i \geq 2$. In exact arithmetic this upper endpoint is nonnegative. The Rayleigh quotient $\ell_\gamma = \|J_\gamma^T v_0\|^2 \leq \mu_1$ is at least one because the bias is present. Dividing by it and imposing $\rho_i \leq 1$ proves Eq. (A.6).

The residual $r_n = J_\gamma w_n - y$ satisfies

$$r_{n+1} = (I - \eta K_\gamma) r_n, \quad r_0 = -y, \quad \frac{\|r_n\|^2}{\|y\|^2} = \sum_i p_i (1 - \eta \mu_i)^{2n}. \quad (\text{A.17})$$

At $\eta = 1/(2\mu_1)$ the factors are $(1 - \rho_i/2)^{2n}$, decreasing in $\rho_i \in [0, 1]$. Replacing ρ_i by its upper bound and summing against $p_i \geq 0$ proves Eq. (A.7). This step is a fixed conservative fraction of the stability limit $2/\mu_1$, not the maximal stable step. For any nonoscillatory step $0 < \eta\mu_1 \leq 1$, the proof replaces $\rho/2$ by $\eta\mu_1\rho$.

Only energy in the exact nullspace contributes a permanent floor $p_0 = \sum_{\mu_i=0} p_i$. A positive mode is representable because $u_i = J_\gamma(J_\gamma^T u_i / \mu_i)$, even when its decay is slow. The weights p_i use the actual finite-kernel eigenvectors; replacing them with continuous-center eigenvectors would require another argument.

A.4 NUMERICAL EVALUATION AND SOURCES OF SLACK

The analytic statement uses exact H_γ and S_γ . Numerical evaluation uses a positive center-quadrature factor A and an exterior factor C , forming $AA^T - CC^T$. It does not truncate a Fourier series: quadrature evaluates the same integral whose Fourier identity explains the mechanism.

Truncating the integral at $\pm(\max_a |z_a| + P/\gamma)$ omits a positive matrix with norm at most $2e^{-4P}/(h\gamma)$, added to the numerical upper endpoint. The exterior tail enters the lower side of the kernel enclosure. For $Z = [A, C]$, its truncated SVD Z_r , and $D = \text{diag}(I, -I)$, expansion of $Z = Z_r + E$ gives

$$\|ZDZ^T - Z_r D Z_r^T\| \leq 2\sigma_1(Z)\sigma_{r+1}(Z) + \sigma_{r+1}(Z)^2. \quad (\text{A.18})$$

This allowance enters both endpoints. Pad the compressed spectrum with zeros between its positive and negative eigenvalues to recover dimension $m - 1$.

Quadrature and arithmetic error are separate from these analytic tails. Quadrature refinement and sensitivity checks support a numerical evaluation; they are not rigorous error certificates. A multiple of machine epsilon is an empirical guard unless justified by an arithmetic-error proof. Clipping a negative computed upper endpoint at zero cannot repair a violated enclosure. Computed curves should therefore be identified as checked floating-point evaluations of the theorem.

Omitting numerically unresolved target energy weakens the lower curve conservatively. Report unresolved spectral energy separately from the directly computed projection residual; neither alone

proves an exact nonrepresentability floor in finite precision. The analytic bound also loses sharpness through the interlacing index shift and the use of ℓ_γ instead of μ_1 . Additive lattice and numerical allowances can dominate very small eigenvalues. Retaining more exterior centers or improving arithmetic reduces some of this slack, but does not remove the interlacing or normalization gaps. These are distinct from inaccurate numerical evaluation.

All conclusions concern fixed features and the specified sample norm. Population error, evolving centers or slopes, and Adam require additional analysis or empirical evidence.

B Empirical validation and protocol

The frozen-feature experiment tests the GD output-error bound under its stated assumptions. The joint-training experiment asks whether GD and Adam acquire features that permit accurate output within the measured budget. They share the target, width, sample grids, precision, and error metric; the frozen-GD theorem does not apply to evolving features or to Adam.

B.1 FROZEN-FEATURE VALIDATION

The five-million-update comparison tests the theorem against executed GD at every update. At bandwidths $1/32, 1/16, 1/8, 1/4$, the smallest ratios of predicted lower error to actual error are respectively 0.9632, 0.7544, 0.8275, 0.8471. At the final update they are 0.9985, 0.9516, 0.9058, 0.9308. The maximum apparent lower-bound violation is 1.1×10^{-15} ; the independently evaluated exact-spectrum error differs from executed GD by at most 2.9×10^{-14} . Refining the center quadrature changes the predicted error by at most 9.6×10^{-11} . These checks support the plotted floating-point calculation over the tested horizons and geometries.

Attainability and optimization are separate. At $\lambda = 1/32$, a directly fitted readout has dense-grid relative error 6.24×10^{-10} , whereas GD retains error 0.210 after five million updates. At $\lambda = 1/4$, the same GD protocol reaches 3.88×10^{-4} . The direct fits retain singular values above 10^{-12} times the largest singular value, and the very accurate small-slope fit has readout ℓ_1 norm about 7.5×10^3 . It witnesses attainability without claiming that the corresponding coefficients are easy to learn.

The slow error has the expected frequency content. Projecting the executed residual onto the three orthogonal target harmonics shows that $\sin(14\pi x)$ accounts for 95.41% of remaining residual energy at $\lambda = 1/32$ after five million updates. At its frequency, the feature multiplier is $M_\gamma(14\pi) = 0.001463$ for $\lambda = 1/32$ and 0.799774 for $1/4$. These values describe feature smoothing, not eigenvalue ratios. The finite-kernel theorem accounts for the geometry connecting smoothing to GD rates. At larger bandwidths, the late residual lies predominantly outside the three input harmonics, illustrating why the full finite kernel remains relevant even for a target with only three frequencies.

The three recipes highlighted in Figure 5 are cosine Adam at 0.002, constant Adam at 0.001, and cosine Adam at 10^{-4} , respectively. The constant-rate trajectory remains noisy, so its final error is not a sustained level. The full sweep is not monotone: $\lambda = 1$ is worse than $1/2$, consistent with the target-dependent tradeoff in the construction. The reference bandwidth $1/4$ was fixed before this comparison. It is a useful supplied geometry, not a fitted universal optimum or a threshold that applies to heterogeneous learned features.

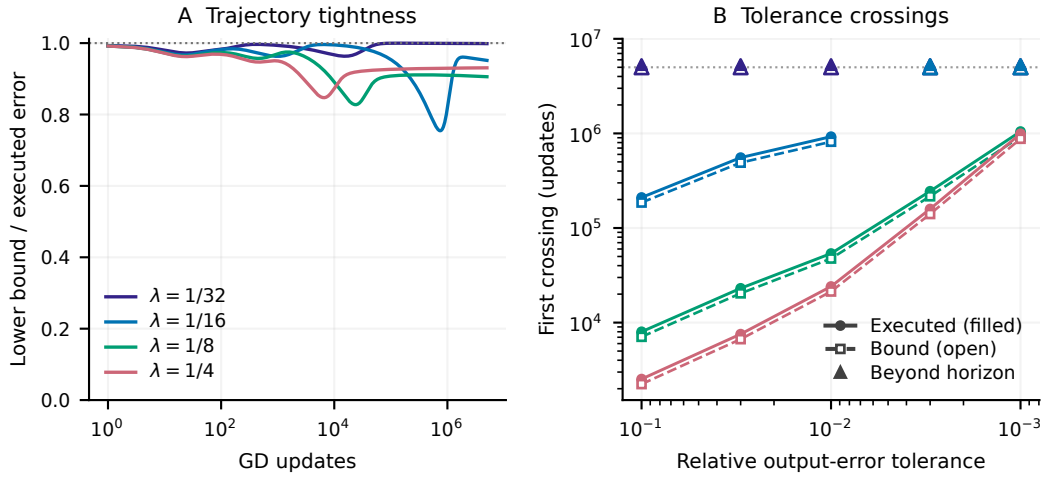


Figure 4: **Theorem validation over five million updates.** Left: predicted lower error divided by executed GD error; one denotes equality. Display curves use a mixed logarithmic/linear grid, while the reported extrema inspect every update. Right: first crossings of several output-error tolerances, with necessary times from the theorem shown open and executed crossings filled. Triangles indicate no crossing within the horizon, not estimated crossing times. Among uncensored comparisons, necessary times are 88.35–88.42% of executed times. No tolerance defines circuit usability in this experiment.

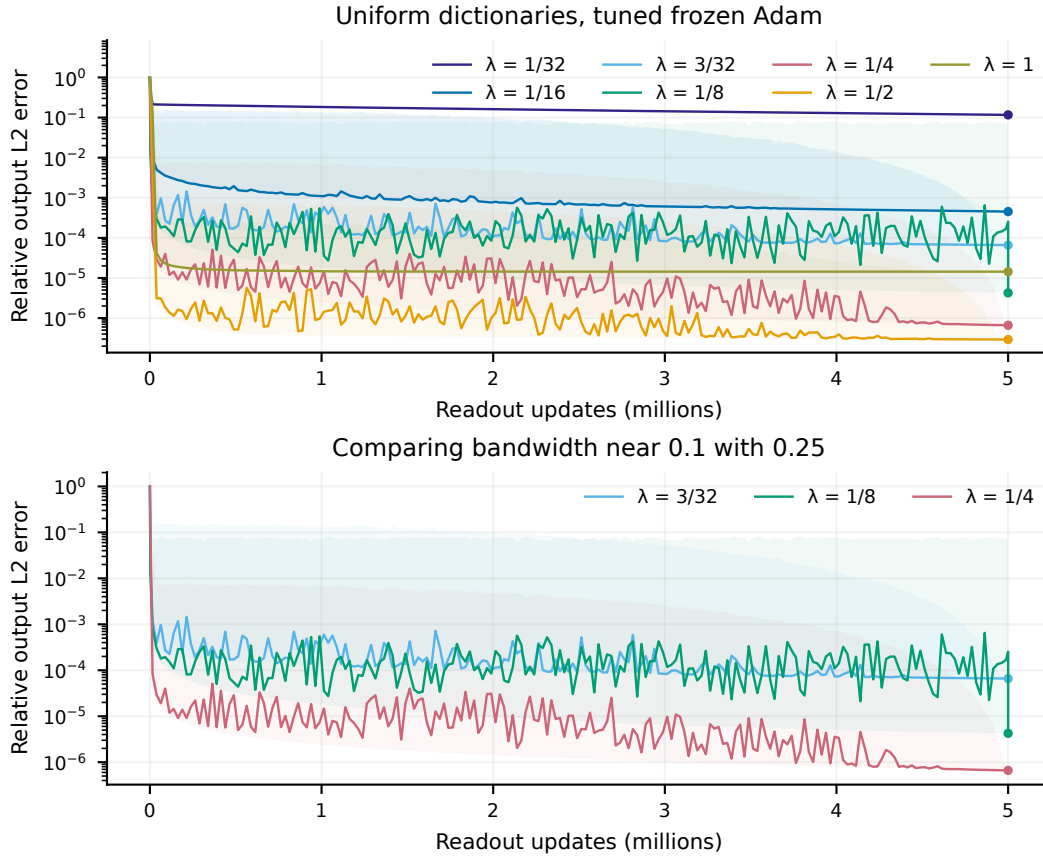


Figure 5: **Frozen Adam under the same five-million-update budget.** Each curve uses one complete recipe chosen by final validation error from the shared expanded search. Lines aggregate compact bin medians into 200 display intervals; shading preserves every within-bin extremum, and dots mark exact endpoint errors. The lower panel isolates $\lambda = 3/32, 1/8, 1/4$, whose final dense-grid errors are 6.63×10^{-5} , 4.26×10^{-6} , and 6.62×10^{-7} . Thus $\lambda = 3/32 \approx 0.094$ leaves about 100 times the error of $1/4$, while $1/8$ leaves 6.44 times its error. These are empirical Adam trajectories, not evaluations of the GD bound.

B.2 CONTROLS ON THE ACQUIRED CENTER-SLOPE GEOMETRY

These controls distinguish readout optimization from changes to the features themselves. They use the primary mixed-sine target, total width 512, five paired seeds, and the grids and error metric in Appendix B. A source optimizer is the optimizer that learned the features; *every iterative follow-up readout uses Adam*, including readouts of GD-acquired features.

Protocol. Readout restarts freeze the features and run two million additional updates from zero readout and zero moments. Each dictionary tests rates $\{10^{-5}/3, 10^{-5}, 10^{-4}, 10^{-3}, 0.002, 0.01, 0.1, 0.3\}$ under constant and full-horizon cosine schedules. One complete recipe is selected per dictionary by final validation error; no earlier checkpoint replaces the endpoint. These additional updates diagnose feature quality rather than matching end-to-end training cost. Direct FP64 SVD fits use a relative singular-value cutoff of 10^{-12} , with 10^{-14} and 10^{-16} as sensitivity checks. Reported errors are recomputed on the 8,192-point dense grid. Independent columnwise reconstruction verifies the saved final fits; on this machine it changes accumulation order, not precision. No truncated fit is interpreted as a certified capacity floor.

Readout access improves as features are acquired. For Adam-acquired features, median restart error decreases from 0.808 at initialization to 0.0408 at 200k, 0.0305 at 600k, and 0.00250 at two million joint updates. For GD-acquired features, the corresponding errors are 0.808, 0.276, 0.256, and 0.251. At the final Adam features, restart error 0.002501 nearly matches the original joint endpoint 0.002509. The direct fit reaches 0.002407, with median coefficient L^2 norm 1.12×10^7 ; cutoff 10^{-14} gives 0.002385 and norm 8.22×10^8 . These fits give little evidence that optimizing this particular frozen readout more accurately would close the gap to the supplied uniform dictionary. For final GD features, the direct-fit median is 0.217, below the restart’s 0.251 but still far from high precision.

Steepening can reduce accuracy. Multiplying both a_j and b_j in $\tanh(a_j x + b_j)$ by four or sixteen preserves its center $-b_j/a_j$. For final Adam features, these changes increase median restart error to 0.1056 and 0.1652, respectively; median paired increases are $42.2\times$ and $66.0\times$. Direct fits yield the same medians at cutoffs 10^{-12} and 10^{-14} , with much smaller coefficient norms than the unscaled fits. For final GD features, restart medians increase to 0.280 and 0.292, although direct-fit medians decrease to 0.205 and 0.156. A slope change therefore affects both the approximation and access to it. The new dictionary need not contain the old span: steeper features can lose a useful approximation. This is not a monotone intervention proving that larger slopes always improve a learned model.

Isolated geometry replacements do not restore high precision. Two further controls use the final unscaled dictionaries. One places their exact signed slopes on the uniform reference centers, with one predeclared permutation per source optimizer and seed. The other preserves learned centers and signs but sets every slope magnitude to $\gamma = 0.25/h$. Neither adds neurons. Their direct-fit medians are respectively 0.1146 and 0.07885 for Adam-acquired features, and 0.3735 and 0.2659 for GD-acquired features. At cutoff 10^{-14} , Adam’s medians are unchanged; GD’s are 0.3384 and 0.2651. Actual matched Adam readout training reaches medians 0.1422 and 0.07929 for Adam-acquired sources, and 0.4734 and 0.2724 for GD-acquired sources. Neither isolated replacement restores the original Adam geometry’s accuracy. Six of twenty selected recipes lie at the already expanded rate-grid boundary; this limits claims about optimal tuning. No further expansion is performed.

Although 97.85–99.02% of Adam’s learned centers lie inside the interval, the 512 centers occupy only 41–49 distinct locations after rounding to 12 decimal places. Most coalesced centers belong to nearly zero-slope units. Interior units with $h|a_j| \geq 0.01$ have 35–37 distinct centers, occupying 27–29 of 32 equal domain bins. Their coverage is much better than unweighted center quantiles suggest. GD has only 5–7 such interior units, occupying 3–7 bins. These thresholds describe the

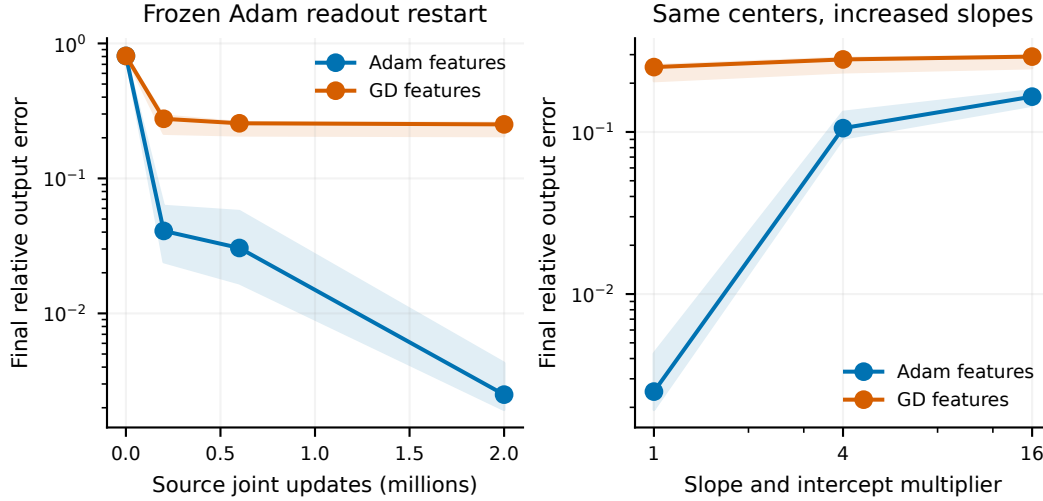


Figure 6: Readout access through acquired features. Source labels identify the joint optimizer; all follow-up readouts use Adam for two million additional updates. Feature snapshots and final slope interventions use the same candidate recipes and final-validation selection rule. The target is the primary mixed sine, with five paired seeds.

measured geometry; they do not define a theorem assumption or an invariant active-neuron count. Assigning a common large slope reactivates many nearly coincident features and need not supply 512 distinct useful directions.

The controls support interpreting the joint-training gap through the acquired center-slope geometry, rather than a single slope statistic. The uniform-dictionary theorem quantifies slope-dependent readout delay under its stated geometry; these controls neither extend it to arbitrary joint training nor prove an Adam convergence law.

Acquisition continues over the longer horizon. The independently checked five-million-update Adam comparison selects cosine decay from rate 0.05 and reaches median dense error 0.001815, a 27.7% reduction from 0.002509 at two million updates. These are fresh runs with decay over their respective full horizons; not every seed improves. Its seeds have 50–56 neurons with $h|a_j| \geq 0.01$, versus 36–43 at two million updates, and 41–45 with $h|a_j| \geq 0.1$, versus 25–29. The 99th-percentile slopes reach 0.233–0.257, but only 4–7 of 512 neurons reach 0.25. An upper percentile near the uniform reference does not mean the full width acquired a common slope. These counts show continued acquisition without establishing a required count for arbitrary geometries. Appreciably sloped interior centers occupy 28–31 of 32 bins. For the remaining units, raw center-coalescence counts can be numerically misleading: three seeds contain 421–457 subnormal input slopes, making center ratios of nearly vanished features delicate. The earlier interventions remain controls on the two-million-update source dictionaries.

Repeating the readout assays on the actual five-million-update source features gives median error 0.00181474 for Adam-acquired features and 0.24209 for GD-acquired features, compared with 0.002501 and 0.25134 from the two-million-update source features. Every assay still uses *two million additional Adam readout updates*, regardless of source optimizer or horizon. At the final Adam features, the restarted readout nearly reproduces the original joint error 0.00181482.

Separate interventions on the five-million-update source features also worsen accuracy. Multipliers four and sixteen give median errors 0.09161 and 0.13893 for Adam-acquired features, with median paired increases of 41.7× and 69.2× over their unmodified restarts. Every seed worsens. GD-acquired features give 0.27952 and 0.28804, also worse in every seed. The adverse intervention result thus persists at the longer source horizon.

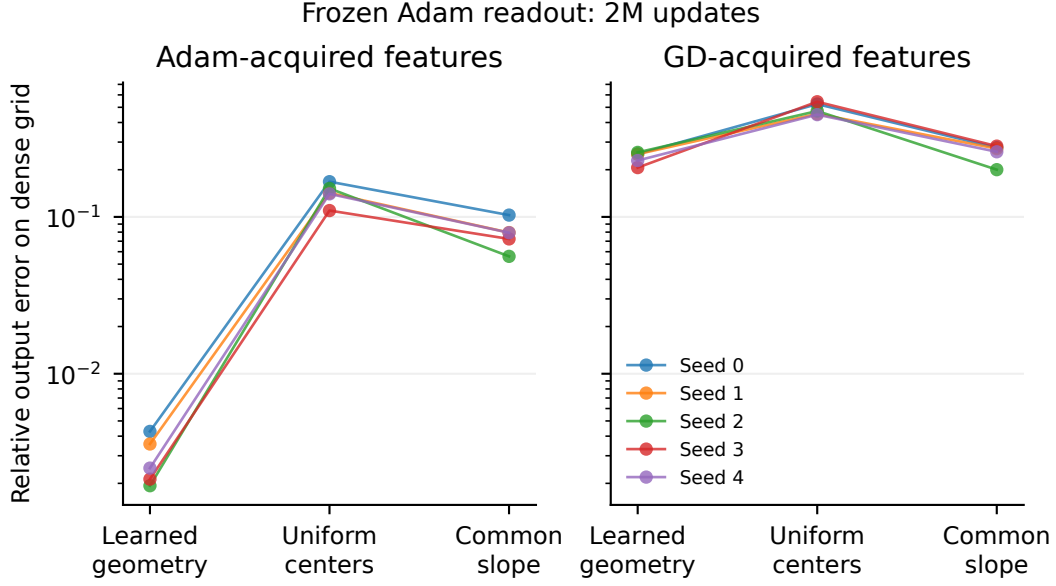


Figure 7: Executed geometry-replacement controls, paired by seed. Categories show the original learned geometry, uniform centers with its learned slopes, and its learned centers with common slope magnitude $0.25/h$. Panels distinguish the source joint optimizer. Every dictionary, including the original baseline, receives two million additional frozen Adam updates after two million joint updates; final validation error selects a complete recipe. The vertical axis is relative error on the dense resolution-check grid. Neither replacement improves median accuracy.

Direct fits to the final five-million-update Adam features reach median dense error 0.0018137 at cutoff 10^{-12} , compared with actual joint error 0.0018148 , with median coefficient norm 1.89×10^6 . Cutoff 10^{-14} gives 0.0018119 and norm 3.10×10^8 ; the 10^{-16} cutoff does not improve the median. The longer horizon improves both the actual output and the accuracy found by these direct fits, while leaving little measured gain from refitting the final readout. This is not a lower bound on the error of all possible readouts.

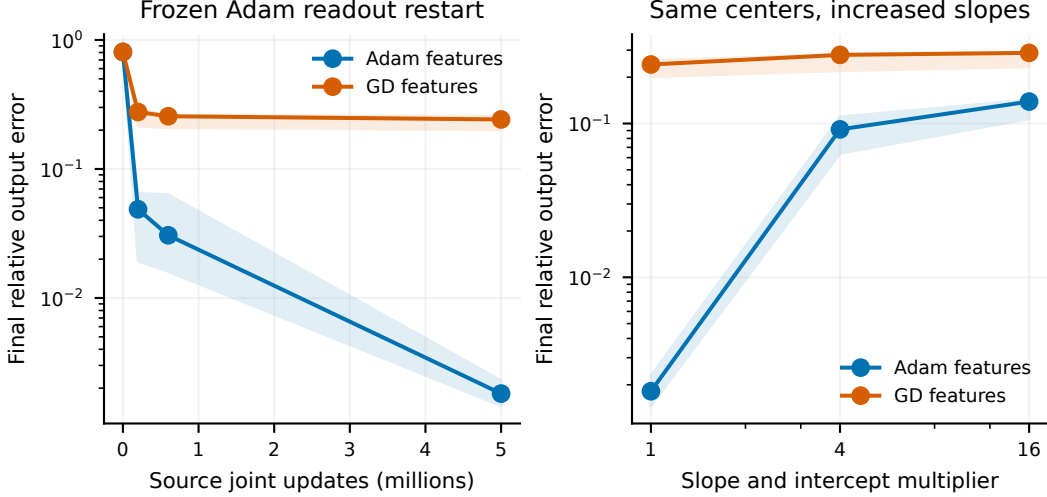


Figure 8: Five-million-update source trajectories tested with two million additional frozen Adam updates per dictionary. Left: median dense error from features acquired at the indicated source update, with the five-seed range shaded. Right: the same readout protocol after multiplying slopes and intercepts of the final five-million-update features. Legends identify the source optimizer; all readout restarts use Adam. Earlier snapshots belong to these fresh full-horizon schedules, not to the previous two-million-update source runs.

B.3 TARGET DEPENDENCE: A QUADRATIC CONTROL

Replacing the mixed sine by x^2 , normalized by its training-grid RMS, tests whether the acquired slope scale depends on target content. We retain width 512, five paired initializations, FP64 arithmetic, and the grids and output-error criterion in Appendix B. Following the two-million-update rate search and a lower-rate Adam expansion, one rate per optimizer and schedule is rerun from initialization for five million updates. Cosine decay spans this full horizon. This secondary control retains one rate per schedule, whereas the primary target retains two. Final validation error selects one complete recipe per optimizer by its median across the five seeds; no earlier checkpoint or per-seed recipe replaces it. The 8,192-point grid checks resolution rather than providing an untouched generalization test.

Adam selects cosine decay from 0.002 and reaches median dense relative error 5.434×10^{-5} ; GD selects constant rate 0.1 and reaches 6.298×10^{-4} . Their median RMS values of $h|a_j|$ are 0.001357 and 0.0005944, and their median neuronwise 99th percentiles are 0.007401 and 0.001503, with $h = 2/467$. This smoother target therefore reaches lower error with substantially smaller slope statistics than the mixed sine. The uniform-dictionary reference $1/4$ is not a universal threshold for learned heterogeneous features.

The longer horizon reduces Adam’s median error from 1.633×10^{-4} at two million updates, with all five seeds improving. Its trailing-window median raw error still decreases by 35.7% across the final fifth of the five-million horizon, while the 99th-percentile slope decreases by 1.23%. Thus error improvement need not track a monotone increase in a scalar slope statistic, and these finite-budget results do not establish permanent training failure.

Separate frozen-dictionary checks at two million updates. At $\lambda = 1/32, 1/16, 1/8, 1/4$, the theorem lower bound divided by executed GD error stays above 0.7633, 0.7827, 0.7852, 0.7852 at every update; its endpoint ratios are 0.8453, 0.9117, 0.9324, 0.9347. The largest apparent violation is 2.0×10^{-15} , and the exact-spectrum reference differs from executed GD by at most 8.5×10^{-15} . Among uncensored tolerance crossings, the necessary times are 62.4–88.0% of executed times. These are checked floating-point evaluations, not interval certificates. The separate frozen Adam sweep

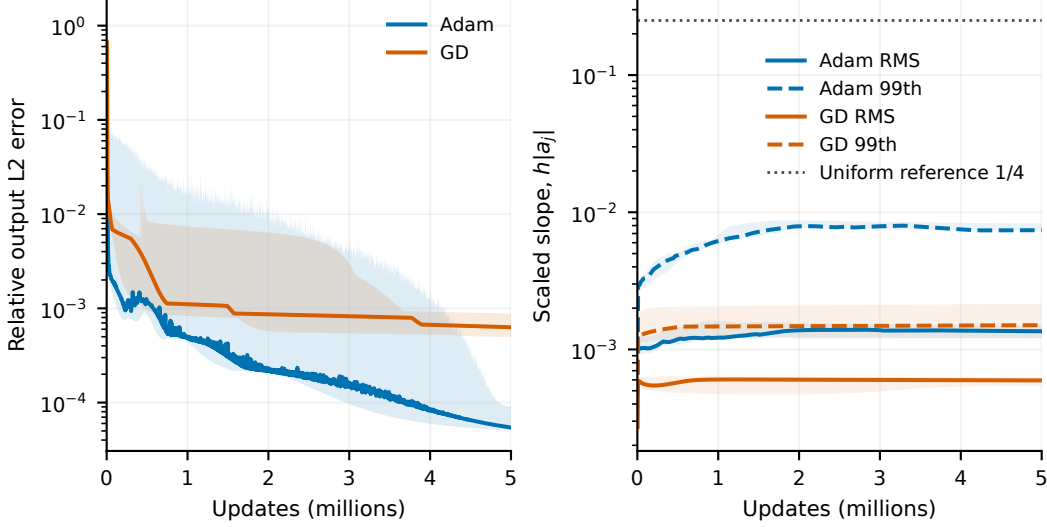


Figure 9: Quadratic joint training over five million updates, using five paired seeds. Left: raw relative training error, with median lines and shading retaining seed ranges and within-bin extrema. Right: RMS and neuronwise 99th-percentile scaled slopes from the same runs; shading shows seed ranges and also preserves within-bin extrema for RMS. The dotted line is a uniform-dictionary reference, not a required slope for this target.

reaches dense errors 2.328×10^{-4} , 1.399×10^{-4} , and 5.985×10^{-5} at $\lambda = 3/32, 1/8, 1/4$. Every selected recipe is at a boundary of the expanded rate grid; these are best-in-grid outcomes, not optimizer optima. Their two-million-update budget is kept separate from the joint figure.

B.4 EXPERIMENTAL PROTOCOL

Targets, width, and sampling. The primary target on $[-1, 1]$ is

$$f(x) = \frac{\sin(2\pi x) + \frac{1}{2} \sin(6\pi x) + \frac{1}{4} \sin(14\pi x)}{\sqrt{21/32}}. \quad (\text{B.1})$$

Its components contain $16/21$, $4/21$, and $1/21$ of its squared norm. A supporting experiment uses x^2 , normalized by its training-grid RMS. Neither target is generated by a small-slope network. All models have $W = 512$ hidden units and an output bias. Uniform frozen dictionaries allocate this *total* width as $N = 467$ intervals and $R = 22$ halo centers on each side: $h = 2/467$, $x_j = -1 + jh$ for $j = -22, \dots, 489$, and $W = N + 2R + 1$. Their common slope is $\gamma = \lambda/h$. Joint models train all parameters in $q(x) = d + \sum_{j=1}^{512} c_j \tanh(a_j x + b_j)$.

Training, validation, and sampling-resolution checks use respectively 2,048, 4,096, and 8,192 equally spaced midpoints of $[-1, 1]$. Validation selects optimizer recipes; the denser grid checks sampling sensitivity and is not an untouched test set. For a grid X , every reported relative output error is

$$e_X(q) = \frac{\|q(X) - f(X)\|_2}{\|f(X)\|_2}. \quad (\text{B.2})$$

Training is full-batch FP64 minimization of $\|q(X) - f(X)\|_2^2 / (2|X|)$. The main trajectories show raw relative error, without an exponential moving average or a tolerance-based definition of success.

Initialization and optimizer selection. Frozen readouts and their output biases start at zero, as do Adam’s moments. For joint training, five paired seeds draw input slopes, hidden intercepts, and

independently drawn readout coefficients uniformly from $[-\sqrt{6/513}, \sqrt{6/513}]$; the output bias is zero. The initial parameter arrays are shared across optimizers, schedules, and targets. The initial horizon is $H = 2 \times 10^6$ updates. Adam tests rates $\{0.0002, 0.002, 0.02, 0.05, 0.1\}$ and GD tests $\{0.0002, 0.002, 0.02, 0.05, 0.1, 0.2, 0.5, 1\}$. Every rate uses both a constant schedule and the full-horizon cosine schedule

$$\eta_n = \frac{\eta_0}{2} \left[1 + \cos \left(\frac{\pi n}{H} \right) \right], \quad 0 \leq n < H. \quad (\text{B.3})$$

Adam uses $(\beta_1, \beta_2, \epsilon) = (0.9, 0.999, 10^{-8})$. A winning boundary rate receives one factor-three outward expansion; divergent recipes remain in the record.

For each optimizer, one complete recipe is selected by median final validation error across all five seeds, requiring finite endpoints in every seed. We neither choose recipes separately by seed nor splice recipes over time. The reported endpoint is the final checkpoint, not an earlier minimum. The selected joint-error and slope trajectories always refer to the same runs.

Duration checks. The best constant-rate recipe is checked separately from the selected schedule. A greater-than-10% error reduction or 99th-percentile slope increase over the final fifth triggers a five-million-update follow-up with the best two rates per optimizer and schedule. Error reduction compares trailing windows of length $H/100$ ending at $0.8H$ and H . This allocates computation; it is not a convergence criterion. Every longer cosine run starts again from its paired initialization with decay over its new horizon.

For the primary target, neither optimizer met the constant-run trigger at two million updates. A discretionary five-million-update duration check was nevertheless executed because the selected Adam cosine trajectory still reduced its trailing-window error by 20.60% over the final fifth. The longer comparison uses Adam constant rates 0.02, 0.002, Adam cosine rates 0.05, 0.02, and GD rates 0.2, 0.5 under both schedules, restarting from the paired initialization. The selected five-million-update recipes are cosine Adam at 0.05 and cosine GD at 0.2. Adam’s median dense-grid error falls from 0.0025095 at two million updates to 0.0018148 at five million, although one of the five paired seeds worsens. Its five-million-update cosine trajectory still reduces trailing-window error by 28.90% over the final fifth; the best constant-rate trajectory reduces it by 18.42%. These are finite-budget comparisons, not converged optimizer optima. In the quadratic experiment, Adam’s constant rate 0.0002 *did* meet the duration trigger: its error reduction was 30.82%. Its lower-boundary expansion to 0.0002/3 was worse under both schedules, leaving the winning rates interior to the expanded grid. The supporting quadratic extension advances the best rate per schedule across all five seeds, rather than the two rates used for the primary extension. This budget adjustment follows measured small-batch throughput. Its five-million-update joint curves are compared over the same horizon; the quadratic uniform readouts were executed for two million updates and are reported separately.

Slope statistics and trajectory preservation. Every raw joint-training error and RMS slope is saved; parameters are saved every 10,000 updates. The displayed slope statistics are the RMS and 99th percentile of $h|a_j|$. Here h is the fixed reference spacing; jointly learned centers need not be uniform. A horizontal common-slope reference therefore does not assert a necessary threshold for heterogeneous features. Plot reduction preserves the entire horizon, exact endpoints, and within-bin extrema, so short instability is not removed by downsampling.

Theorem bounds and exact-spectrum checks. Frozen GD uses bandwidths $1/32, 1/16, 1/8$, and $1/4$, zero readout, and $\eta = 1/(2\mu_1)$, where μ_1 is the largest eigenvalue of $K = JJ^T$ and J is the feature matrix, including its bias column, divided by $\sqrt{m}$. Write μ_i, u_i for the kernel eigenpairs, $\rho_i = \mu_i/\mu_1$, and $p_i = |u_i^T f(X)|^2 / \|f(X)\|_2^2$. The theorem supplies upper rate bounds $\bar{\rho}_i(\gamma)$, clipped to $[0, 1]$ where appropriate. Its predicted lower curve is

$$e_{\text{lower}}(n) = \left[\sum_{i \in \mathcal{I}_{\text{resolved}}} p_i \left(1 - \frac{\bar{\rho}_i(\gamma)}{2} \right)^{2n} \right]^{1/2}. \quad (\text{B.4})$$

Here $\mathcal{I}_{\text{resolved}}$ retains computed ratios exceeding 10^{-18} . The plotted lower curve omits numerically unresolved modes and the unresolved projection remainder rather than treating them as a certified capacity floor. This omission only weakens a lower bound. Target projections use the actual dictionary. The gamma-dependent upper rate bounds, rather than its actual small eigenvalues, enter this prediction. Using the computed ρ_i , all computed target weights, and the projection remainder gives the separately identified exact-spectrum GD check. Predictions are evaluated before reading executed training trajectories, and the lower curve is compared with saved GD error at every update. Two center-quadrature resolutions check numerical sensitivity. These are checked floating-point evaluations of the analytic theorem, not directed-rounding numerical certificates.

Uniform readout baselines and attainable error. Frozen Adam tests bandwidths $1/32, 1/16, 3/32, 1/8, 1/4, 1/2$, and 1 with rates $\{10^{-5}, 10^{-4}, 10^{-3}, 0.002, 0.01, 0.1\}$ and both schedules. Boundary winners prompted a shared expansion by $10^{-5}/3$ and 0.3 ; the expanded rates apply to every uniform dictionary and learned-feature assay. The preselected main reference is $\lambda = 1/4$, using one complete recipe chosen by final validation error. The completed five-million-update uniform comparison uses the complete expanded grid.

A direct SVD readout fit independently checks attainable accuracy. The reference fit retains singular values exceeding 10^{-12} times the largest singular value; cutoffs 10^{-14} and 10^{-16} provide sensitivity checks. Output errors are recomputed from the fitted coefficients on all three grids, and coefficient norms are retained. A truncated-SVD residual is not a capacity floor, and the cutoff is not selected separately at each slope to minimize displayed error.

Learned-feature and adaptive-metric diagnostics. Readout-restart assays freeze selected joint features at initialization, 200,000, 600,000, and the final update, then reset the readout and its optimizer moments. Separate final-feature interventions multiply both a_j and b_j by four or sixteen, preserving each nonconstant feature’s center. These interventions test slope changes at learned centers and do not impose the uniform-center assumptions of the theorem. Each assay has a common two-million-update readout budget; this is additional diagnostic computation, not a matched end-to-end training budget.

For a kernel or adaptive metric A , with eigenvalues ν_i and normalized eigenvectors z_i , the energy below relative cutoff τ is reported as

$$\mathcal{E}_A(v; \tau) = \frac{\sum_{\nu_i/\nu_1 \leq \tau} |z_i^T v|^2}{\|f(X)\|_2^2}, \quad v \in \{f(X), f(X) - q(X)\}. \quad (\text{B.5})$$

Thus target and residual energies use the same denominator. Adam’s endpoint second moments define the adaptive metric

$$A = J \operatorname{diag}[(\sqrt{\hat{v}} + \epsilon)^{-1}] J^T, \quad (\text{B.6})$$

where $\hat{v}$ is the bias-corrected second-moment vector of the actual run. Omitting the scalar learning rate leaves relative eigenvalues defined even at the end of cosine decay. This is an endpoint diagnostic of adaptive coordinate scaling, not an Adam convergence law: its momentum and changing second moments are not captured by a fixed metric.

At actual joint-training endpoints, the readout Jacobian contains $\tanh(a_j x + b_j)$ and the bias column. The full Jacobian additionally contains $x c_j \operatorname{sech}^2(a_j x + b_j)$ and $c_j \operatorname{sech}^2(a_j x + b_j)$, for slopes and hidden intercepts. Both Jacobians are divided by $\sqrt{m}$; their adaptive metrics use the saved moments of the joint Adam run, restricted to the relevant parameter blocks. Moments from a restarted readout are not substituted for joint-training moments. Selected recipes, continuation decisions, and figure provenance remain attached to their run records. Finite-budget error is reported as such and does not establish permanent optimization failure.

Constant-rate control for the adaptive-metric interpretation. At the five-million-update endpoints in Figure 3, the full Jacobian with the actual Adam metric places at least 99.9997% of each seed’s remaining residual energy below relative eigenvalue 10^{-6} . Figure 10 shows the target and residual projections before and after adaptive scaling. The near-flat readout residual curves mean that almost all remaining error is already confined to the weakest readout directions; the full Jacobian supplies additional directions, but their relative eigenvalues are also small.

At the selected two-million-update cosine Adam endpoints, over 99.998% of residual energy lies below relative eigenvalue 10^{-6} in the full Jacobian metric with actual Adam scaling. The best constant-rate recipe, $\eta = 0.02$, gives median 98.763%, with seed range 47.671–99.309%. These fractions divide by the endpoint residual energy; the cumulative curves themselves divide by target energy. Thus weak local directions also matter without cosine cooldown, although the concentration is not uniform across constant-run seeds.

The constant runs’ median adaptive-trace-weighted contribution of ϵ to $\sqrt{\hat{v}} + \epsilon$ is 7.71×10^{-6} for the full Jacobian and 1.09×10^{-6} for its readout block, versus approximately 24% and 23% at the cosine endpoints. Counting coordinates with $\sqrt{\hat{v}} \leq \epsilon$ exaggerates its effect because many have negligible Jacobian columns. The constant-rate check therefore supports the weak-direction interpretation without relying on an epsilon-dominated metric; it remains a local diagnostic, not an Adam convergence theorem.

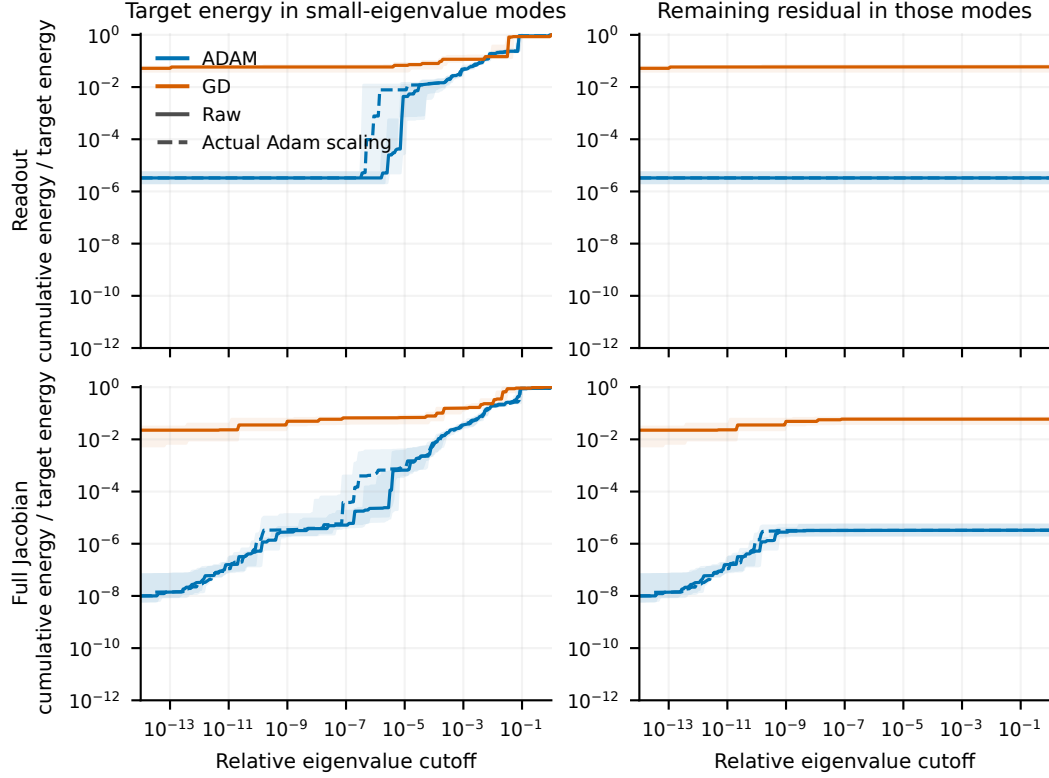


Figure 10: **Remaining error occupies weak local directions.** Five-million-update endpoints of the selected joint runs. Each curve sums target or residual energy below the horizontal relative-eigenvalue cutoff, divided by target energy; its right endpoint is therefore one or the squared relative output error. Rows use readout and full-parameter Jacobians. Dashed blue curves apply the saved joint Adam second-moment scaling. Lines show seed medians and shading shows ranges. These endpoint metrics diagnose geometry; they do not predict Adam’s trajectory.

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